

IMAT MOCK EXAM 11: Last Rehearsal

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 42/90.0 · Mock exam

Section A - Reading Skills, General Knowledge & Logical Reasoning (Q1-9)

1. "Take this medicine at evenly spaced times as prescribed and complete the full course, even if your symptoms improve, because stopping early can let the infection return. If you forget a dose, take it as soon as you remember, but skip the missed dose if your next dose is due soon; never take a double dose. Return any unused tablets to a pharmacy: they must not be kept to treat future illnesses or given to other people."

From the informational leaflet for an antibiotic

Which of the following statements is the correct interpretation of the instructions?

- A) Stopping the course when symptoms improve is the recommended way to avoid side effects.
- B) Leftover tablets must not be saved to treat a later illness.
- C) Two tablets should be taken together whenever a dose is missed.
- D) This antibiotic should always be taken with food.
- E) Sharing the medicine with a family member who has similar symptoms is acceptable.

2. "In a large Spanish trial, volunteers at high cardiovascular risk were assigned by lottery to either a Mediterranean-style diet supplemented with extra-virgin olive oil and nuts, or to a control diet advised only to limit fat. Over about five years, the Mediterranean groups suffered roughly 30% fewer heart attacks, strokes and deaths from cardiovascular causes than the control group. The benefit appeared without meaningful weight loss and without a rise in calorie intake. The researchers note that the diet as a whole was tested, not any single ingredient, and that participants lived in a Mediterranean country where these foods are familiar and affordable."

From a newspaper report on nutrition research

Indicate which of the following statements is not correct:

- A) The benefit did not depend on the participants losing weight.
- B) Major cardiovascular events were about 30% lower in the Mediterranean groups.
- C) The researchers evaluated a whole dietary pattern rather than one ingredient.
- D) The trial demonstrated the protective effect of extra-virgin olive oil alone.
- E) Assignment to the Mediterranean or the control diet in the trial was random.

3. "With the telescope I have detected four small stars that have escaped the notice of every astronomer before me. They attend a certain planet, arranged in a straight line with it, and their distances from the planet change: at times one hides behind the planet or passes in front of it, and they complete their circuits in unequal periods, the nearest the fastest."

Adapted from Galileo Galilei, Sidereus Nuncius (1610)

What did the author observe?

- A) Four moons orbiting a planet
- B) Spots on the surface of the Sun
- C) Comets passing close to the Sun
- D) Fixed stars that never change position
- E) The rings of Saturn

4. "An antibiotic kills susceptible bacteria but spares any cells that already carry resistance genes, for example a gene encoding an enzyme that destroys the drug. The surviving cells multiply, and resistance genes can also pass between bacteria on small circles of DNA. In environments where antibiotics are used often, such as hospitals, resistant strains therefore make up a growing share of the bacterial population over time."

From a microbiology textbook

Only one of the following statements is coherent with the excerpt:

- A) Antibiotics create resistance genes inside susceptible bacteria.
- B) Susceptible and resistant cells are killed by an antibiotic to the same degree.
- C) Bacteria deliberately mutate when an antibiotic is present so that they can survive it.
- D) Resistance genes can never move from one bacterium to another.
- E) Frequent use of antibiotics favours the spread of resistant strains.

5. Consider this statement: All squares are rectangles.

Which of the following statements is deducible from the given statement?

- A) All rectangles are squares.
- B) Some rectangles are not squares.
- C) If a figure is not a rectangle, then it is not a square.
- D) Every four-sided figure is a rectangle.
- E) Squares and rectangles share no geometric properties.

6. From the given statements:

If it rains this afternoon, the tennis match will be moved indoors.

If the match is moved indoors, it will start at 2 pm.

The match did not start at 2 pm.

Which conclusion can be deduced?

- A) The match started late.
- B) It did not rain this afternoon.
- C) The match was cancelled.
- D) The match was moved indoors but started at a different time.
- E) It rained this afternoon.

7. Amara, Bruno and Chiara live on three different floors of a small building, one person per floor. Bruno lives on the third floor, Amara lives directly above Bruno, and Chiara lives directly above Amara. On which floor does Chiara live?

- A) On the sixth floor.
- B) On the fourth floor.
- C) On the third floor.
- D) On the second floor.
- E) On the fifth floor.

8. Three students, Sara, Tom and Leo, each study exactly one foreign language among French, German and Spanish. Sara studies neither French nor German, and Leo studies German. Which of the following is correct?

- A) Tom studies French.
- B) Tom studies German.
- C) Tom studies Spanish.
- D) Sara studies French.
- E) Leo studies Spanish.

9. What is the angle between the hour hand and the minute hand of an analogue clock at exactly 9:30?

- A) 75°
- B) 90°
- C) 105°
- D) 120°
- E) 180°

Section B - Biology (Q10-32)

- 10.** The small amount of ATP that glycolysis itself produces is formed by:
- A)** oxidation of pyruvate to acetyl-CoA.
 - B)** hydrolysis of the NADH produced in the pathway.
 - C)** chemiosmosis at the inner mitochondrial membrane.
 - D)** direct transfer of a phosphate group from a phosphorylated intermediate to ADP.
 - E)** the electron transport chain.
- 11.** The reaction that converts pyruvate into acetyl-CoA before entry into the Krebs cycle takes place in:
- A)** the cytosol.
 - B)** the intermembrane space.
 - C)** the Golgi apparatus.
 - D)** the inner mitochondrial membrane.
 - E)** the mitochondrial matrix.
- 12.** During the complete oxidation of one glucose molecule to CO₂ and water, carbon dioxide is released during:
- A)** glycolysis and oxidative phosphorylation.
 - B)** glycolysis and the Krebs cycle.
 - C)** the link reaction and the Krebs cycle.
 - D)** the Krebs cycle and oxidative phosphorylation.
 - E)** glycolysis and the link reaction.
- 13.** A substance makes the inner mitochondrial membrane leaky to protons while leaving the electron transport complexes untouched. The most direct consequences are that:
- A)** electron transport stops while ATP synthesis continues.
 - B)** electron transport accelerates while ATP synthesis falls.
 - C)** ATP synthesis increases while electron transport slows.
 - D)** the proton gradient grows and ATP synthesis accelerates.
 - E)** electron transport and ATP synthesis both stop.
- 14.** In the mitochondrial respiratory chain, electrons are carried from complex III to complex IV by:
- A)** cytochrome c. **B)** ubiquinone. **C)** oxygen. **D)** NAD⁺. **E)** ATP synthase.
- 15.** In haemoglobin, each oxygen molecule binds to:
- A)** a molecule of bicarbonate inside the cell.
 - B)** the membrane of the red blood cell.
 - C)** a terminal amino group of a globin chain.
 - D)** the iron ion at the centre of a heme group.
 - E)** a hydrophobic pocket between two subunits.
- 16.** The largest fraction of the carbon dioxide carried in venous blood is transported:
- A)** as bicarbonate ions in the plasma.
 - B)** inside the organelles of red blood cells.
 - C)** as carbamino compounds on plasma proteins.
 - D)** dissolved directly in the plasma.
 - E)** bound to the iron atoms of haemoglobin.
- 17.** Compared with fast-twitch fibres, slow-twitch (type I) muscle fibres:
- A)** contain fewer mitochondria and fatigue rapidly.
 - B)** rely mainly on anaerobic glycolysis for their ATP.
 - C)** contain more mitochondria and are more resistant to fatigue.
 - D)** contract more rapidly but for shorter periods.
 - E)** store no myoglobin at all.

- 18.** The thin layer of cells lining the inner surface of the heart chambers and of all blood vessels is called the:
- A)** mesothelium.
 - B)** endothelium.
 - C)** periosteum.
 - D)** collagen sheath.
 - E)** perichondrium.
- 19.** The activity of an enzyme rises with temperature up to an optimum and then falls steeply at higher temperatures because the enzyme:
- A)** switches to catalysing a different reaction.
 - B)** has run out of substrate.
 - C)** is broken down into its component fatty acids.
 - D)** starts to synthesise an inhibitor of itself.
 - E)** begins to denature, losing its functional three-dimensional shape.
- 20.** Inhibition by a non-competitive inhibitor differs from competitive inhibition in that it:
- A)** affects only the apparent affinity of the enzyme for its substrate.
 - B)** is reversed completely by adding excess substrate.
 - C)** disappears when the reaction mixture is cooled.
 - D)** lowers the maximum rate of the reaction and is not overcome by excess substrate.
 - E)** involves the inhibitor binding at the active site.
- 21.** Between meals, bile is stored and concentrated in the:
- A)** liver.
 - B)** stomach.
 - C)** small intestine.
 - D)** pancreas.
 - E)** gall bladder.
- 22.** Soon after a meal rich in carbohydrates, the liver responds to the prevailing hormonal signal by:
- A)** converting glucose into glycogen for storage.
 - B)** converting excess glucose into urea.
 - C)** breaking glycogen down into free glucose.
 - D)** increasing the loss of glucose in the bile.
 - E)** secreting glucagon into the blood.
- 23.** The osmotic gradient of the renal medulla, which allows the production of concentrated urine, is created mainly by active salt transport in the:
- A)** proximal convoluted tubule.
 - B)** distal convoluted tubule.
 - C)** loop of Henle.
 - D)** glomerulus.
 - E)** renal artery.
- 24.** The cells whose beating cilia sweep mucus and trapped dust upward out of the airways are:
- A)** alveolar macrophages.
 - B)** ciliated epithelial cells.
 - C)** chondrocytes.
 - D)** type I alveolar cells.
 - E)** goblet cells.

- 25.** Iodine is required in the human diet mainly for the synthesis of:
- A)** cortisol.
 - B)** adrenaline.
 - C)** antidiuretic hormone.
 - D)** thyroxine.
 - E)** insulin.
- 26.** The α -helix and the β -pleated sheet are features of which level of protein structure?
- A)** Primary structure
 - B)** Secondary structure
 - C)** The prosthetic group level
 - D)** Tertiary structure
 - E)** Quaternary structure
- 27.** According to the fluid mosaic model of the plasma membrane, the proteins of the membrane:
- A)** can drift laterally within the lipid bilayer.
 - B)** are composed of carbohydrate.
 - C)** form a solid continuous coat over the bilayer.
 - D)** occur only on the outer surface of the membrane.
 - E)** are each fixed rigidly in a single position.
- 28.** The digestive enzymes inside lysosomes have an optimum pH of about:
- A)** 2. **B)** 5. **C)** 7.4. **D)** 9. **E)** 12.
- 29.** Catalase, the enzyme that decomposes hydrogen peroxide into water and oxygen, is concentrated in which organelle?
- A)** The Golgi apparatus
 - B)** The mitochondrion
 - C)** The nucleolus
 - D)** The lysosome
 - E)** The peroxisome
- 30.** For a cross between two heterozygotes, $Aa \times Aa$, the expected ratio of genotypes $AA : Aa : aa$ among the offspring is:
- A)** 1:1:1 **B)** 1:2:1 **C)** 2:1:1 **D)** 3:1 **E)** 9:3:3:1
- 31.** Red-green colour blindness is caused by a recessive allele carried on the X chromosome. A man with colour blindness and a woman with normal vision whose father was colour blind have children. Which statement is correct?
- A)** All of the children will have normal vision.
 - B)** All of the daughters will be colour blind.
 - C)** On average, half of the sons and half of the daughters will be colour blind.
 - D)** All of the sons will be colour blind.
 - E)** The daughters cannot have received the allele from their father.
- 32.** During elongation of translation, the new peptide bond forms between the growing polypeptide chain and:
- A)** the anticodon of the following tRNA.
 - B)** the ribosomal RNA of the small subunit.
 - C)** the tRNA about to leave through the E site.
 - D)** the amino acid carried by the tRNA that has just entered the A site.
 - E)** the codon currently being read in the mRNA.

Section C - Chemistry (Q33-47)

- 33.** What is the pH of a 0.001 mol/dm^3 aqueous solution of sodium hydroxide, a strong base? ($K_w = 1.0 \times 10^{-14}$ at 25°C)
A) 3.0 **B)** 7.0 **C)** 10.0 **D)** 11.0 **E)** 13.0
- 34.** At 25°C , pure water has $K_w = 1.0 \times 10^{-14}$ and pH 7. At 80°C , K_w rises to 4.0×10^{-13} . In pure water at 80°C , the concentration of H^+ and the nature of the water are:
A) $[\text{H}^+] = 1.0 \times 10^{-7} \text{ mol/dm}^3$ and the water is neutral.
B) $[\text{H}^+] = 6.3 \times 10^{-7} \text{ mol/dm}^3$ and the water is acidic.
C) $[\text{H}^+] = 6.3 \times 10^{-7} \text{ mol/dm}^3$ and the water is neutral.
D) $[\text{H}^+] = 2.0 \times 10^{-13} \text{ mol/dm}^3$ and the water is basic.
E) $[\text{H}^+] = 4.0 \times 10^{-13} \text{ mol/dm}^3$ and the water is acidic.
- 35.** Which of the following substances has a giant covalent (network covalent) structure?
A) Diamond
B) Sodium chloride
C) Methane
D) Ice
E) Copper
- 36.** Water ($M = 18 \text{ g/mol}$) boils at 100°C , while methane ($M = 16 \text{ g/mol}$) boils at -162°C . The main reason for the much higher boiling point of water is:
A) ionic bonds between water molecules.
B) the larger number of covalent bonds inside each water molecule.
C) the slightly greater molecular mass of water.
D) covalent bonds linking the atoms of different molecules.
E) hydrogen bonds between water molecules.
- 37.** What is the oxidation number of phosphorus in H_3PO_4 ?
A) +1 **B)** +3 **C)** +5 **D)** +6 **E)** +7
- 38.** In the reaction $2\text{FeCl}_2 + \text{Cl}_2 \rightarrow 2\text{FeCl}_3$, which species is the oxidising agent?
A) Fe^{2+} , because it combines with chlorine.
B) Fe^{3+} , because it is the product of the reaction.
C) Cl^- , because it carries a negative charge.
D) Neither species, because no electrons are transferred.
E) Cl_2 , because it accepts electrons from the iron and is itself reduced.
- 39.** How many open-chain alcohols have the molecular formula $\text{C}_4\text{H}_{10}\text{O}$?
A) 2 **B)** 3 **C)** 4 **D)** 5 **E)** 6
- 40.** The bond formed between the amino group of one amino acid and the carboxyl group of another is called:
A) a phosphodiester bond.
B) an ester bond.
C) a glycosidic bond.
D) a peptide bond.
E) a hydrogen bond.
- 41.** What amount, in moles, is present in 8.8 g of carbon dioxide? ($M(\text{CO}_2) = 44 \text{ g/mol}$)
A) 0.1 mol **B)** 0.2 mol **C)** 0.44 mol **D)** 2.0 mol **E)** 4.4 mol
- 42.** A sample of gas occupies 4.0 dm^3 at 300 K and 100 kPa . If the sample is heated to 600 K at constant pressure, its volume becomes:
A) 2.0 dm^3 **B)** 4.0 dm^3 **C)** 6.0 dm^3 **D)** 8.0 dm^3 **E)** 12.0 dm^3

43. Going down group 1 of the periodic table, from lithium towards francium, the reactivity of the metals with water:

- A)** increases, because the outer electron is further from the nucleus and is lost more easily.
- B)** remains constant, because every group 1 metal has one outer electron.
- C)** decreases, because the outer electron is held more tightly.
- D)** decreases, because the atoms become smaller.
- E)** first increases and then decreases.

44. The electron configuration of a neutral atom of sulfur ($Z = 16$) is:

- A)** $1s^2 2s^2 2p^6 3s^2 3p^4$
- B)** $1s^2 2s^2 2p^6 3s^2 3p^6$
- C)** $1s^2 2s^2 2p^6 3s^1 3p^5$
- D)** $1s^2 2s^2 2p^5 3s^2 3p^5$
- E)** $1s^2 2s^2 2p^4 3s^2 3p^6$

45. For the equilibrium $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, the equilibrium constant K is given by:

- A)** $K = 2[SO_3] / (2[SO_2] + [O_2])$
- B)** $K = ([SO_2]^2[O_2]) / [SO_3]^2$
- C)** $K = [SO_3] / ([SO_2]^2[O_2])$
- D)** $K = [SO_3]^2 / ([SO_2][O_2])$
- E)** $K = [SO_3]^2 / ([SO_2]^2[O_2])$

46. When the temperature of a reaction mixture is raised, the rate of the reaction increases. Which property of the reaction remains unchanged?

- A)** The rate at which products form.
- B)** The number of reactant molecules able to react.
- C)** The activation energy of the reaction.
- D)** The frequency of collisions between reactant molecules.
- E)** The fraction of molecular collisions energetic enough to react.

47. The compound formed between the aluminium ion, Al^{3+} , and the sulfate ion, SO_4^{2-} , has the formula:

- A)** Al_2SO_4 **B)** $Al_2(SO_4)_3$ **C)** $Al(SO_4)_3$ **D)** $Al_3(SO_4)_2$ **E)** $AlSO_4$

Section D - Mathematics (Q48-54)

48. Solve the inequality $1 - 2x \geq 5$.

- A)** $x \leq -2$ **B)** $x \geq 2$ **C)** $x \geq -2$ **D)** $x \leq -1$ **E)** $x \leq 2$

49. The set of real solutions of the inequality $(x - 2)^2 \geq 0$ is:

- A)** $x \geq 2$
- B)** only $x = 2$
- C)** the empty set
- D)** the set of all real numbers
- E)** $x \leq 2$

50. In a class of 40 students, 25 study French, 18 study German and 9 study both languages. How many students study neither language?

- A)** 6 **B)** 9 **C)** 15 **D)** 16 **E)** 34

51. If $3x + 2y = 12$ and $y = 3$, then x equals:

- A)** 1 **B)** 2 **C)** 3 **D)** 4 **E)** 6

52. The expression $x^2 - 16$ can be factorised as:

- A)** $(x - 4)^2$ **B)** $(x + 4)^2$ **C)** $x(x - 16)$ **D)** $(x - 2)(x + 8)$ **E)** $(x - 4)(x + 4)$

53. A right-angled triangle has legs of 6 cm and 8 cm. Its perimeter is:

- A)** 14 cm **B)** 20 cm **C)** 24 cm **D)** 26 cm **E)** 30 cm

54. In a right-angled triangle, one angle is 30° and the hypotenuse has length 10 cm. The side opposite the 30° angle has length:

- A)** 2.5 cm **B)** 5 cm **C)** $5\sqrt{3}$ cm **D)** 10 cm **E)** 12 cm

Section E - Physics (Q55-60)

55. The velocity-time graph of a cyclist is a horizontal line at 6 m/s for the whole interval from $t = 0$ to $t = 10$ s. The distance covered in this interval is:

- A)** 0.6 m **B)** 6 m **C)** 30 m **D)** 60 m **E)** 600 m

56. When a 12 V battery is connected across a lamp, a current of 0.5 A flows. The resistance of the lamp is:

- A)** 6 Ω **B)** 12 Ω **C)** 24 Ω **D)** 48 Ω **E)** 60 Ω

57. A 60 W lamp is left switched on for 2 hours. The electrical energy it consumes is:

- A)** 0.012 kWh **B)** 0.12 kWh **C)** 1.2 kWh **D)** 12 kWh **E)** 120 kWh

58. A box slides across a level floor at constant velocity. The resultant (net) force acting on the box is:

- A)** directed forwards and larger than the friction force.
B) equal to the friction force and directed forwards.
C) equal to the weight of the box.
D) directed backwards.
E) zero.

59. An astronaut weighs about one sixth as much standing on the Moon as standing on Earth because:

- A)** the Moon's smaller gravitational acceleration acts on the astronaut's unchanged mass.
B) the astronaut's body contains less matter on the Moon.
C) the Moon has no atmosphere to push up on the astronaut.
D) weight cannot be measured where there is no water or air.
E) part of the astronaut's weight is still carried by the distant Earth.

60. An object weighs 50 N in air. When it is fully submerged in water, its apparent weight is 40 N. The buoyant force exerted by the water on the object is:

- A)** 5 N **B)** 10 N **C)** 40 N **D)** 50 N **E)** 90 N

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	B	2	D	3	A	4	E	5	C
6	B	7	E	8	A	9	C	10	D
11	E	12	C	13	B	14	A	15	D
16	A	17	C	18	B	19	E	20	D
21	E	22	A	23	C	24	B	25	D
26	B	27	A	28	B	29	E	30	B
31	C	32	D	33	D	34	C	35	A
36	E	37	C	38	E	39	C	40	D
41	B	42	D	43	A	44	A	45	E
46	C	47	B	48	A	49	D	50	A
51	B	52	E	53	C	54	B	55	D
56	C	57	B	58	E	59	A	60	B